

LEARNING FROM DATA

MIHIR RAO

Contents

1	The Learning Problem	2
1.1	From observations to a rule	2
1.2	A simple model: the perceptron	2
1.3	The perceptron learning algorithm	3
1.4	Three kinds of feedback	4
1.5	The necessity of assumptions	5

LECTURE 1: THE LEARNING PROBLEM

Machine learning is useful when examples reveal a pattern that is hard to write down as a rule. Movie ratings are one example: past ratings can help predict what a viewer will like, even without a full description of their tastes or every movie's features.

1.1 From observations to a rule

Let $\mathcal{X}$ be the input space and $\mathcal{Y}$ the output space. A credit applicant, for example, may be represented by an input $x \in \mathcal{X}$ containing age, salary, employment history, and debt, while $y \in \mathcal{Y}$ records whether the applicant is a good or bad credit risk.

Definition 1.1 [TARGET FUNCTION]. The target function $f: \mathcal{X} \rightarrow \mathcal{Y}$ is the unknown rule we want to learn. In the credit example, it is the ideal credit approval function. We observe examples of its outputs, not the rule itself.

The learner observes a data set

$$\mathcal{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}, \quad y_n = f(x_n),$$

and returns a FINAL HYPOTHESIS $g: \mathcal{X} \rightarrow \mathcal{Y}$. The goal is not merely to reproduce the recorded outputs, but to make g behave like f on inputs that were not in $\mathcal{D}$.

Two design choices together form the LEARNING MODEL:

- The HYPOTHESIS SET $\mathcal{H} = \{h\}$ is the set of candidate formulas: the rules the learner can choose from.
- The LEARNING ALGORITHM $\mathcal{A}$ uses the data to choose some $g \in \mathcal{H}$.

Data alone do not determine the answer. The hypothesis set limits the rules we consider; the algorithm uses the examples to choose one. [Figure 1](#) shows how the components fit together.

1.2 A simple model: the perceptron

Consider binary classification with labels $y \in \{-1, +1\}$ and an input $x = (x_1, \dots, x_d)$. A linear decision rule assigns a weight w_i to each attribute and compares the weighted score with a threshold. Introduce a constant coordinate $x_0 = 1$ and absorb the negative threshold into w_0 .

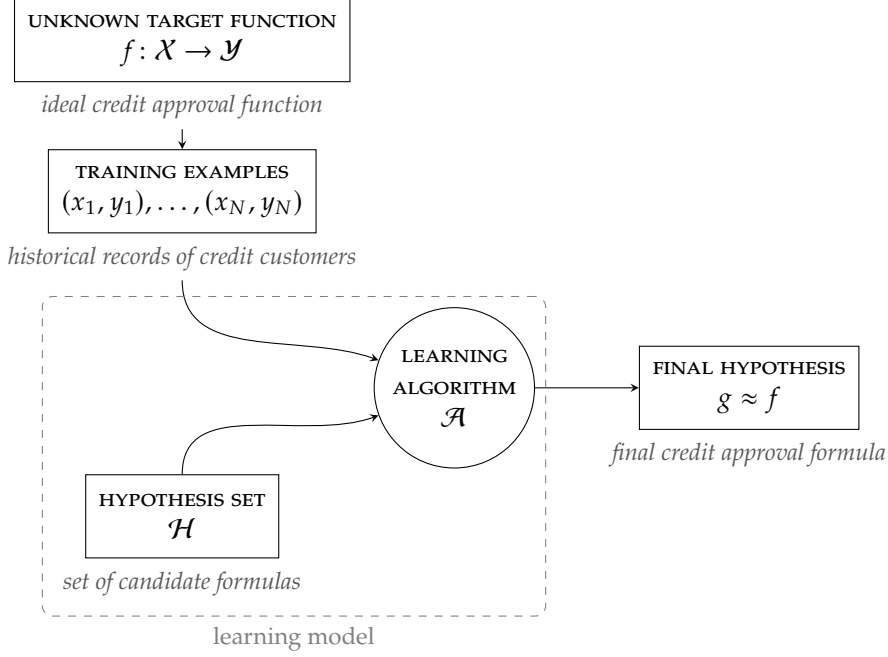


Figure 1: Components of learning. The unknown target f generates the training examples; the learning algorithm uses them to pick the final hypothesis g from the hypothesis set $\mathcal{H}$. The hypothesis set and the algorithm together form the learning model.

With the augmented vectors

$$x = (1, x_1, \dots, x_d) \quad \text{and} \quad w = (w_0, w_1, \dots, w_d),$$

the hypothesis becomes

$$h_w(x) = \text{sign}(w^\top x).$$

The equation $w^\top x = 0$ defines a **DECISION BOUNDARY**. Points on one side receive label +1 and points on the other side receive label -1. A labeled data set is **LINEARLY SEPARABLE** when some w classifies every training point correctly (figure 2, left).

Example: In the credit example, a positive weight can represent an attribute that supports approval, while a negative weight can represent one that counts against it. The bias w_0 sets the baseline needed to cross the approval threshold.

1.3 The perceptron learning algorithm

The **PERCEPTRON LEARNING ALGORITHM** (PLA) searches for a suitable weight vector. Starting from an initial w , it repeatedly selects a misclassified training example (x_n, y_n) satisfying

$$\text{sign}(w^\top x_n) \neq y_n$$

and performs the update

$$w \leftarrow w + y_n x_n.$$

This update has a direct geometric meaning (figure 2, middle and right). For $y_n = +1$ it adds x_n to w , turning w toward x_n ; for $y_n = -1$ it subtracts x_n , turning w away from x_n . Its effect on the selected point's signed score is

$$y_n(w + y_n x_n)^T x_n = y_n w^T x_n + \|x_n\|^2.$$

The positive second term moves that example toward the correct side of the decision boundary. PLA alternates between finding an error and making this correction; when no training point is misclassified, the current w defines the final hypothesis.

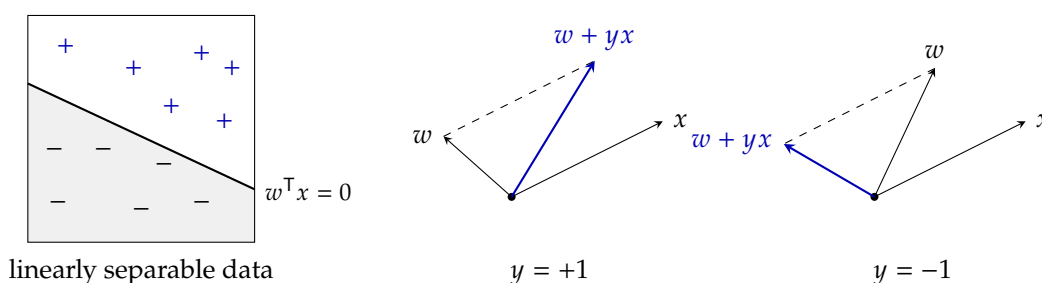


Figure 2: Left: a linearly separable data set; the line $w^T x = 0$ separates the $+1$ points from the -1 points. Middle and right: one PLA update on a misclassified point x . When $y = +1$ the angle between w and x exceeds 90° , and $w + x$ turns toward x . When $y = -1$ the angle is below 90° , and $w - x$ turns away from x .

Note. PLA specifies both parts of the model: linear separators form the hypothesis set, and the iterative correction rule is the learning algorithm. Whether the result also performs well beyond the training data is a separate question.

1.4 Three kinds of feedback

Learning settings differ in the feedback the learner receives.

Definition: In SUPERVISED LEARNING, every training input is paired with a desired output. Coin recognition is an example: measurements such as mass and size come with the coin denomination, allowing the learner to infer decision boundaries (figure 3, left and middle).

Definition: In UNSUPERVISED LEARNING, each training example is a pair (input, ?) rather than (input, correct output). The learner gets inputs without correct outputs and looks for structure, such as groups of similar examples. Without denominations, the same coin measurements still fall into visible clusters (figure 3, right).

Definition: In REINFORCEMENT LEARNING, the learner chooses an output or action and receives a grade for that choice rather than the correct action itself. The feedback evaluates behavior but does

not directly reveal what should have been done.

These settings differ in the information supplied to the learner, not in the basic objective of extracting useful structure from observations.

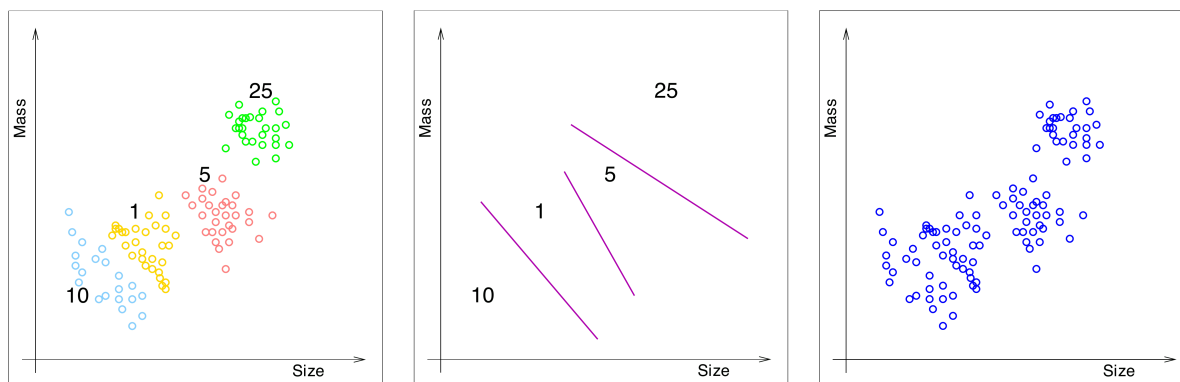


Figure 3: Coin recognition from size and mass. Left: labeled training data, with each coin's denomination (1, 5, 10, or 25 cents) as its output. Middle: decision boundaries learned from the labels. Right: the same measurements without labels, as in unsupervised learning; the clusters remain visible.

1.5 The necessity of assumptions

A small collection of labeled examples may be compatible with many distinct target functions. A visual pattern puzzle (figure 4) can suggest a natural label for a new shape, but another rule can agree with all displayed labels and predict the opposite answer. Every +1 pattern is mirror symmetric and no -1 pattern is, so symmetry labels the new pattern +1. Yet every -1 pattern, and no +1 pattern, has a black top-left cell, so that rule labels it -1. Both rules fit all six examples.

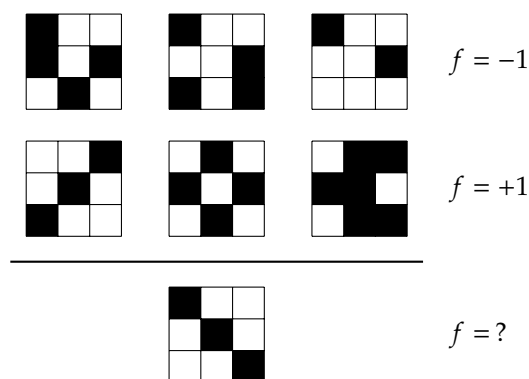


Figure 4: A learning puzzle. Six 3×3 patterns are labeled -1 (top row) or +1 (middle row); the task is to label the new pattern below.

To predict unseen examples, we must make assumptions about which rules are plausible. The choices of $\mathcal{H}$ and $\mathcal{A}$ express those assumptions.

Source: Yaser S. Abu-Mostafa, Learning From Data, Lecture 1, "The Learning Problem," California Institute of Technology, official course page <https://www.work.caltech.edu/telecourse.html> and slides <https://www.work.caltech.edu/slides/slides01.pdf>. Figure 3 is cropped from these slides.